ADS1002_ICU_GROUP3

"What factors most significantly influence patient mortality rates across ICU types 1 through 4?"

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1. Summary

This project investigates the factors that most significantly influence patient mortality across four distinct Intensive Care Unit (ICU) types using predictive modelling and data-driven analysis. The dataset comprises patient demographic, physiological, and biochemical measurements from four specialized ICUs: the Coronary Care Unit (ICU Type 1), the Cardiac Surgery Recovery Unit (ICU Type 2), the Medical ICU (ICU Type 3), and the Surgical ICU (ICU Type 4). Each unit manages different patient populations and clinical conditions, creating unique but interrelated risk profiles for mortality.

The primary objective was to determine which variables most strongly predict in-hospital death within and across these ICU settings. To achieve this, the data were pre-processed, cleaned, and explored through comprehensive exploratory data analysis (EDA), followed by predictive modelling using three machine-learning techniques: Logistic Regression (LogReg), Random Forest (RF), and K-Nearest Neighbors (KNN). Data pre-processing included handling missing values, standardizing feature scales, addressing class imbalance through Synthetic Minority Oversampling (SMOTE), and partitioning the data into training and testing subsets with stratified sampling.

The EDA revealed that neurological, hemodynamic, and metabolic indicators were the most consistent predictors of mortality across all ICUs. Specifically, the Glasgow Coma Scale (GCS), Sequential Organ Failure Assessment (SOFA) score, Blood Urea Nitrogen (BUN), and serum lactate levels demonstrated strong associations with patient death. Unit-specific factors also emerged: Troponin levels were most predictive in cardiac care, postoperative lactate in surgical recovery, and dynamic GCS changes in the medical and surgical ICUs.

Among the models, both Random Forest and Logistic Regression demonstrated strong predictive performance. Random Forest achieved the highest accuracy when sufficient data were available, effectively capturing complex, non-linear relationships between physiological variables. In contrast, Logistic Regression maintained reliable performance even with smaller datasets, offering interpretable insights into the direction and magnitude of predictor effects. KNN provided complementary validation through local similarity detection. Collectively, these models achieved balanced accuracy and recall after threshold optimization, with Random Forest attaining the best overall F1-score on larger datasets.

The findings indicate that ICU mortality arises from the interplay between neurological decline, metabolic acidosis, and hemodynamic instability rather than from any single variable. The study underscores the potential of combining interpretable statistical models with advanced ensemble methods to support early mortality risk detection. These insights can inform targeted interventions, improve clinical decision-making, and ultimately contribute to reducing preventable deaths in critical care settings.

2. Description of project and details of dataset

Description of project

The intensive care unit (ICU) plays a vital role in treating critically ill patients, and ICU mortality is widely used as an indicator of hospital performance and quality of care. However, mortality rates vary widely across ICU types due to differences in patient conditions and treatment contexts. Understanding which factors most strongly affect the mortality rate in each ICU type allows for fair performance comparisons and helps improve patient outcomes.

Thus, for our project, we decided on the main question:
"What factors most significantly affect patient mortality rates across ICU types 1 through 4".

Each group member explored one sub-question focusing on a specific ICU type or overall comparison:

- Sharon: *"What cardiovascular-related factors influence mortality in* ***Coronary Care Unit*** *(ICU Type 1)?"*
- Jaek: *"Which surgical and hemodynamic factors affect mortality in* ***Cardiac Surgery Recovery Unit*** *(ICU Type 2)?"*
- Nafil: *"What metabolic and neurologic markers predict mortality in the* ***Medical ICU*** *(ICU Type 3)?"*
- Min Ge*: "How do post-operative vitals and organ dysfunction metrics affect mortality in* ***Surgical ICU*** *(ICU Type 4)?"*
- Jun: *"Are there common predictors of mortality shared across* ***all ICU types****, or are predictors ICU-specific?"*

The main goal of our project is to build predictive models for each ICU type using the dataset provided, which would allow us to compare the mortality rates between different ICU types and to identify risk factors within and across the different ICU types. The model also aims to predict the probability for survival or non-survival for individual patients based on their clinical data. This is relevant in real life as the models could serve as comparison benchmarks for evaluating ICU performance and supporting clinicians in identifying high risk patients. This would facilitate better resource allocation and target interventions, where more focus should be put on the factors that most strongly affect mortality rate between different types of ICU.

To do this, our group started with (1) data wrangling and cleaning, (2) exploratory data analysis to identify the factors that stood out, and (3) model development and evaluation (including threshold tuning) to obtain the best performing model.

Details of the dataset

The dataset provided to us was sourced from PhysioNet/CinC Challenge 2012.

```
print(df.shape)
```

```
(4000, 394)
```

Figure: Dimension of the provided dataset

The dataset contained 394 variables for 4000 patients, where the variables included 42 descriptors with information about each patient's demographic, physiological, and lab outcomes within their first 48 hours of stay. The data collected was either continuous or categorical, with Boolean being distinguished by 0 and 1. "-1" and "NA" denote missing or unknown data.

Of these variables, 6 of the descriptors are **general descriptors** collected during the patient's admission, which are *RecordID*, *age*, *gender*, *height*, *weight*, and the *ICU type* they were admitted into – Coronary Care (Type 1), Cardiac Surgery Recovery (Type 2), Medical (Type 3), and Surgical (Type 4). The four ICU type variables were one-hot encoded, meaning that the categorical values are transformed into binary columns indicating their category.

The **target variable** we mainly use in the dataset is "*In.hospital_death*", where 0 signifies survival and 1 signifies non-survival. Other important outcome related variables include:

- *SOFA score* (Sequential Organ Failure Assessment score to assess organ dysfunction)
- *SAPS-I score* (Simplified Acute Physiological score used to predict likelihood of patient death)
- *Length of stay* in the ICU
- *Survival* (number of days between ICU admission and death if a patient's death was recorded regardless of whether it was in or out of the hospital).

These outcome related variables can also provide valuable insights regarding both ICU-specific and cross-ICU through predictive modelling.

The rest of the descriptors carry **time series information**, where the frequency of observations may differ between patients (variables can be recorded once, more than once, or none at all). These descriptors provide information about the patient's medical conditions, such as *heart rate*, *cholesterol level*, *GCS* (Glasgow Coma Score which measures patient consciousness), and *respiration rate*. For each descriptor, readings taken within the first 24 hours are labelled with 1 and readings taken between 24 to 48 hours are labelled with 2. The minimum, maximum, and average values are also recorded for all descriptors at both time frames to summarise the variations of the readings over the first day and second day.

Clinically, the dataset reflects real ICU scenarios, making findings potentially applicable to real-world hospital risk assessment.

Relevance of dataset to our project

The dataset is well suited for our research question as it provides detailed clinical and physiological information of patients within the first 48 hours of ICU stay. The number of time series clinical descriptors allows us to identify what variables are most strongly related with mortality in their specific ICU types, for example we would expect heart related predictors to have strong correlation with mortality in the Coronary Care unit (ICU type 1). This would allow us to identify early indicators of mortality for the four ICU types.

3. Preprocessing & Manipulation of Data

Overview

This study uses the PhysioNet 2012 intensive care dataset and prepares admissions from all ICU cohorts for downstream analysis and modelling of in-hospital mortality. Pre-processing was designed to preserve clinically meaningful signals while avoiding information leakage and overfitting. The unit of analysis is a single ICU admission with a binary mortality outcome.

Normalising Missingness and Verifying Structure

The raw data encode missing entries in multiple ways, including explicit nulls and placeholders such as “NA” and the value -1. All such placeholders were treated as missing so that completeness auditing, imputation, and learning algorithms behaved consistently. Variables representing time-window summaries (minimum, maximum, average, and standard deviation) were verified as numeric features suitable for correlation screening and model input. Administrative identifiers were retained for traceability but excluded from predictor sets. After joins across source tables, admissions were checked for duplication and for alignment of survival labels with admission and discharge timing.

Physiological Validity and Data Cleaning

Row-level validity checks were applied to remove values that were impossible or clearly attributable to unit or entry errors while retaining extreme but clinically plausible measurements. Anthropometric fields were examined for unrealistic magnitudes; physiologically impossible vital-sign values and oxygenation settings were excluded; and temporal variables were verified to rule out negative lengths of stay or inconsistent survival timing. When a value could reasonably be considered clinically extreme rather than erroneous, it was retained; removal was reserved for entries that violated physical limits or unit conventions. Domain-specific adjustments were applied where necessary. Glasgow Coma Scale totals were treated as numeric scores and, when required, rounded and constrained to the range from three to fifteen to correct drift introduced by averaging or input anomalies. Binary indicators of mechanical ventilation were harmonised to a consistent coding. Continuous respiratory support measures such as the fraction of inspired oxygen were checked to ensure consistent units within cohorts. Laboratory values were reviewed for unit consistency; values inconsistent with the expected scale were excluded, whereas clinically plausible extremes were preserved to reflect true physiological severity.

Completeness Screening

To reduce instability arising from heavy imputation, completeness was assessed prior to modelling. Counts of valid observations were computed after normalising placeholders to missing and only entries that were neither placeholders nor null were deemed observed. Features were retained for modelling when they met cohort-appropriate completeness thresholds. Variables that failed these thresholds were excluded from modelling matrices to avoid fragile imputations, although they could still be referenced in exploratory summaries where appropriate.

Feature Screening and Redundancy Reduction

Feature selection proceeded in two stages. First, simple associations between each numeric feature and the mortality outcome were computed using Pearson correlation on the training data. A modest absolute correlation threshold was applied to remove clear noise while retaining clinically plausible signals. Second, redundant representations of the same physiological concept were consolidated. Many measurements are recorded as multiple time-window summaries of the same underlying variable and are therefore highly correlated. To mitigate multicollinearity and maintain parsimony, features were grouped into physiological families by their base name, and a single representative variant with the strongest absolute association on the training data was retained from each family. Where necessary to preserve interpretability and sample efficiency, the final design was limited to a fixed number of families.

Imputation and Scaling

After defining the feature set, the model matrix was constructed and missing entries were imputed. Continuous features were imputed using the median, which is robust to the skewed and heavy-tailed distributions typical of intensive care measurements. Binary indicators with rare missingness were imputed using the most frequent category when required. All imputers were fitted on the training split only and then applied unchanged to validation and test data to prevent information leakage. Continuous predictors were standardised to zero mean and unit variance after imputation. Standardisation was necessary for scale-sensitive methods such as logistic regression and k-nearest neighbours and was retained across model families to ensure fair comparison, even though tree-based methods are scale-invariant.

Data Splitting and Leakage Prevention

Stratified sampling was used to create training and test partitions so that the prevalence of the mortality outcome was preserved across splits. All fittable components of the pipeline, including imputation, scaling, correlation screening, family consolidation, and any classification threshold selection, were fitted exclusively on the training data and then applied to held-out partitions. This protocol ensured that no information from the validation or test sets influenced feature selection or parameter estimation.

Rationale and Outcomes

The pre-processing workflow was guided by two principles: preserving genuine physiological signals and preventing bias. Normalising heterogeneous missingness codes ensured that completeness accounting and imputation behaved predictably. Validity checks removed only physically impossible or unit-inconsistent values. Median imputation and standardisation produced numerically stable inputs without distorting heavy-tailed distributions. Family-aware selection reduced redundancy while maintaining one representative of each clinically meaningful concept. The resulting design matrices are clean, compact, and interpretable, and they support fair model comparison while reducing the risk of overfitting.

4. Exploratory Data Analysis

The exploratory data analysis (EDA) provided the foundation for understanding the structure, quality, and clinical significance of the data across the four ICU types. Each ICU represented a distinct patient group, with expected differences in variable distributions and mortality patterns. The analysis examined physiological, biochemical, and demographic relationships to identify potential predictors of mortality for subsequent modelling. Across all ICUs, neurological status, hemodynamic stability, and metabolic function consistently emerged as the strongest indicators of patient outcomes.

ICU Type 1 – Coronary Care Unit (CCU)

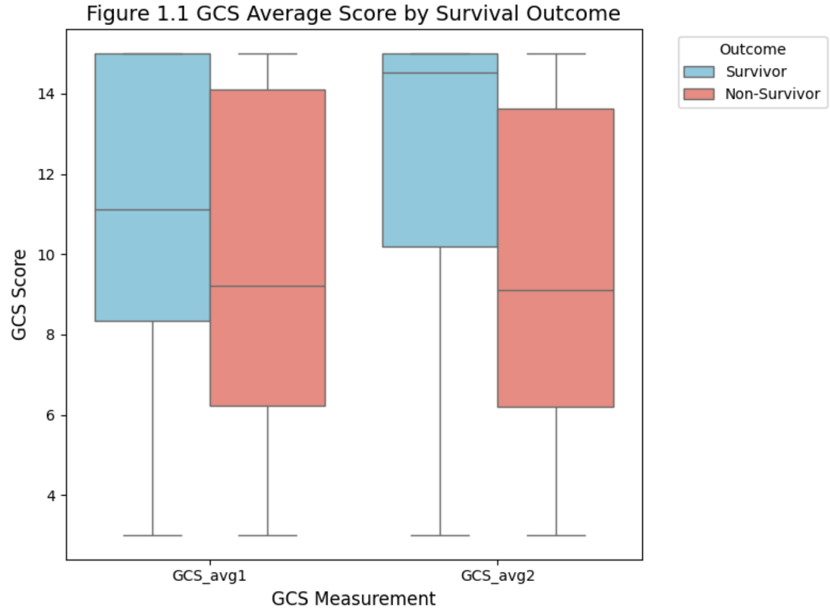

Figure 1.1 : Box plot showing improvement in GCS score between records correlating with chance of survival

The Coronary Care Unit dataset contained 577 cases with a mortality rate of 14.04 percent. Exploratory analysis revealed substantial missing data for cardiac biomarkers such as Troponin, limiting their reliability. Where available, Troponin showed the strongest correlation with mortality, followed closely by the Glasgow Coma Scale (GCS), which measures neurological function. Survivors consistently exhibited higher GCS scores than non-survivors, confirming the strong prognostic value of neurological status in cardiac critical care.

Further analysis identified the Sequential Organ Failure Assessment (SOFA) and Simplified Acute Physiology Score (SAPS-I) as key severity indicators positively associated with mortality. Elevated Blood Urea Nitrogen (BUN) and hepatic enzymes (AST and ALT) were also linked to increased death risk, highlighting renal and hepatic dysfunction as important contributors. Conversely, higher mean arterial and systolic blood pressures were protective. Overall, mortality in ICU Type 1 appears driven by a combination of neurological impairment, multi-organ failure, and hemodynamic instability, with Troponin underscoring the need for consistent cardiac biomarker collection in future predictive studies.

ICU Type 2 – Cardiac Surgery Recovery Unit

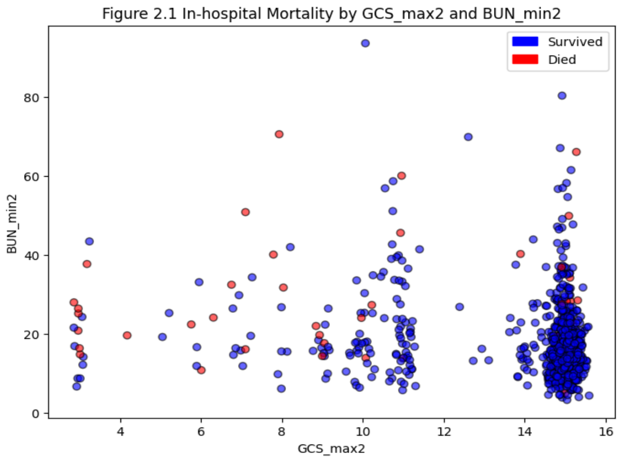

Figure 2.1: Scatter plot showing clustering of survivors at High GCS - Low BUN quadrant

The ICU Type 2 cohort consisted of 874 patients and exhibited the lowest mortality rate across all ICU types, at 4.92 percent. This comparatively low rate reflects the post-surgical but generally stable nature of this population. The exploratory phase focused on two variables that displayed the strongest and most consistent associations with mortality: GCS and BUN. Correlation analyses revealed that GCS_max2 was negatively correlated with in-hospital death, while BUN_max2 was positively correlated. In other words, lower neurological responsiveness and higher kidney dysfunction corresponded to increased mortality risk.

Scatterplot combining GCS and BUN showed a clear compounded risk: patients with both low GCS and high BUN had markedly higher mortality. This interaction highlights the link between neurological and renal function in postoperative recovery. Other metabolic variables such as glucose, PaO_2, and $PaCO_2$ also correlated with death but less consistently due to missing or irregular data. Overall, GCS and BUN emerged as strong individual and combined predictors for identifying post-surgical patients at elevated risk who may benefit from early intervention.

ICU Type 3 – Medical ICU

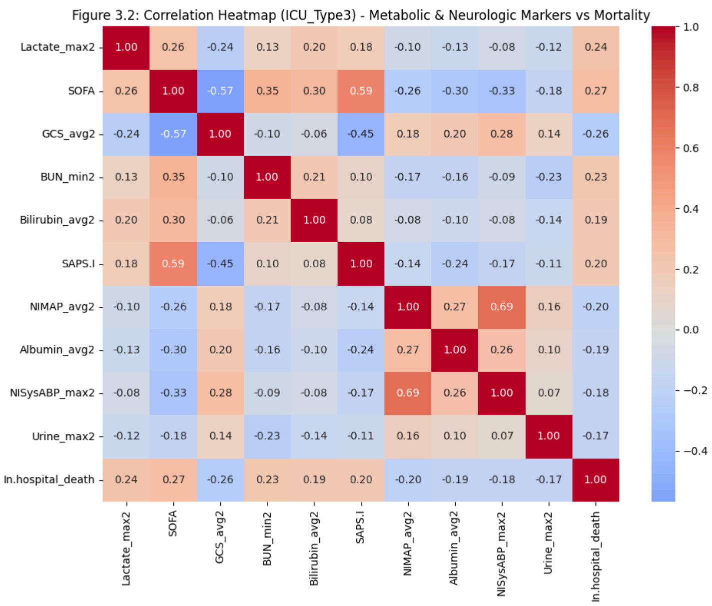

Figure 3.2: Heatmap of chosen variables for predictive modelling, ensuring lack of intercorrelation between predictive variables:

The Medical ICU cohort contained 1,068 records and showed the highest mortality rate at 18.57 percent, reflecting the severity of conditions such as sepsis, respiratory failure, and multi-organ dysfunction. The analysis integrated metabolic, neurologic, and hemodynamic parameters to understand patient deterioration. Correlation analysis identified lactate_max2, SOFA, and GCS_avg2 as the strongest predictors of death. Elevated lactate indicated tissue hypoperfusion and metabolic acidosis, while high SOFA scores reinforced the impact of cumulative organ failure on poor outcomes.

Lower GCS values were again linked to non-survivors, confirming neurological impairment as a universal mortality factor. Additional variables such as BUN, bilirubin, and SAPS-I correlated positively with mortality, while mean arterial pressure and urine output were protective. Heatmaps showed strong clustering among metabolic and organ dysfunction markers. Overall, the Medical ICU analysis revealed that elevated lactate and BUN combined with high SOFA scores signaled multi-organ failure and high mortality risk, establishing SOFA, lactate, and GCS as key early-warning indicators.

ICU Type 4 – Surgical ICU

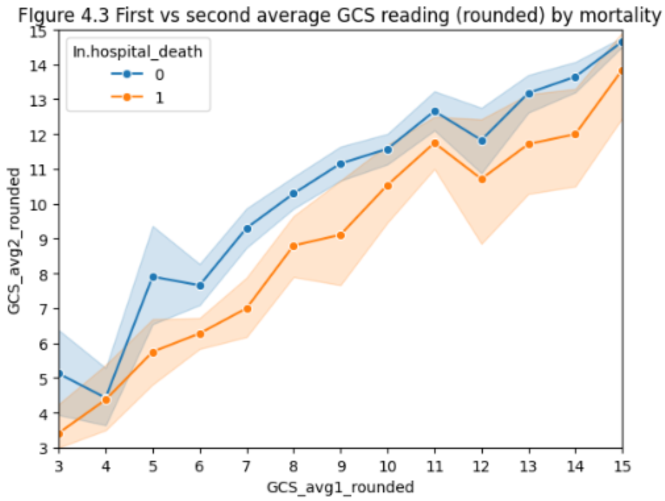

Figure 4.3: line graph showing survivors show greater improvement in mean GCS scores between first and second record

The Surgical ICU dataset included 1,068 patients with a mortality rate of 14.51 percent. Exploratory analysis again highlighted the importance of neurological assessment through the Glasgow Coma Scale (GCS), which showed the strongest correlation with mortality. Although bilirubin also correlated with death, its high proportion of missing values (~60%) limited its use. Visual analysis revealed that patients with lower initial GCS scores were significantly more likely to die.

Sequential GCS measurements uncovered an even clearer pattern: survivors generally maintained or improved their scores over time, whereas non-survivors experienced noticeable declines. Some began with moderate GCS levels but deteriorated sharply before death, emphasizing that changes in GCS over time are more informative than static readings. Other physiological and biochemical measures contributed less consistently, reinforcing the dominance of neurological function as a mortality predictor. These findings suggest that continuous monitoring of GCS trends provides a valuable early-warning signal for postoperative and trauma-related deterioration in surgical ICU patients.

Cross-ICU Exploratory Insights

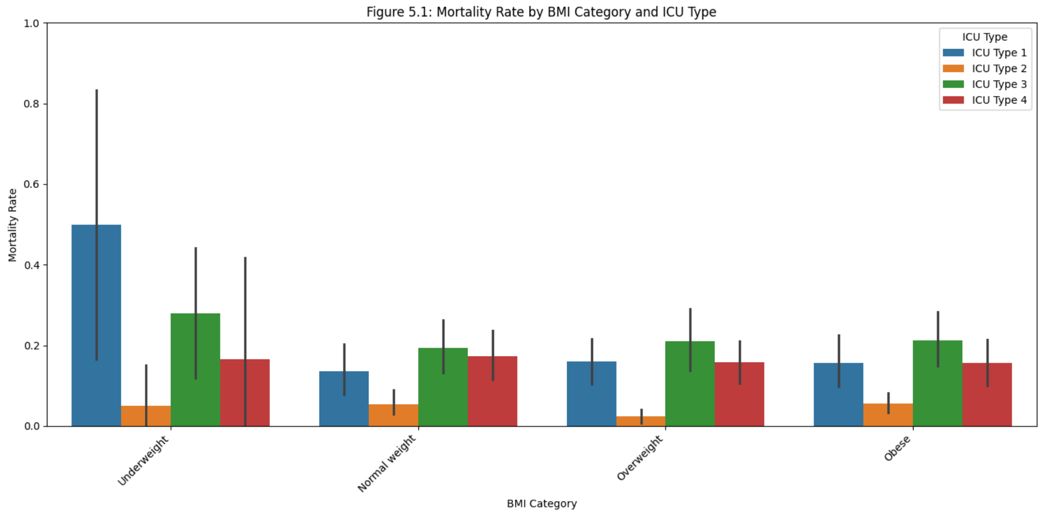

Figure 5.1: Bar chart showing variations of survival rate in each ICU type across BMI groups

Across all four ICU types, consistent patterns emerged. Neurological status, measured by the Glasgow Coma Scale (GCS), was the most reliable predictor of mortality—lower scores consistently indicated higher death risk, while stable or improving scores predicted survival. Organ dysfunction markers such as SOFA, SAPS-I, and BUN were also strongly associated with mortality, reflecting multi-organ failure and renal compromise as key drivers of poor outcomes. Metabolic and circulatory indicators, including lactate, mean arterial pressure, and oxygenation (FiO_2), further highlighted the roles of impaired perfusion and metabolic stress in critical deterioration.

Several ICU-specific patterns were also observed: Troponin and BUN were most significant in cardiac units, while lactate, SOFA, and dynamic GCS changes dominated in medical and surgical units. Additionally, Body Mass Index (BMI) showed a modest but notable association with mortality, suggesting that extreme underweight or obesity may influence recovery and complication risk. Overall, ICU mortality appears to stem from the interaction of neurological decline, metabolic imbalance, hemodynamic instability, and patient-specific physiological factors such as BMI.

Significant Conclusions from the Exploratory Analysis

The exploratory phase provided key insights that shaped model design and variable selection. Across all ICU types, the Glasgow Coma Scale (GCS) consistently emerged as the strongest and most complete predictor of mortality, reflecting both neurological and systemic health. SOFA and SAPS-I scores confirmed that cumulative organ failure strongly influences survival, while renal and hepatic markers such as BUN and AST highlighted the impact of multi-organ stress.

Although biomarkers like Troponin and Bilirubin showed strong correlations in specific ICUs, their limited availability reduced their broader utility, emphasizing the need for more complete clinical data. The analysis also revealed that variable combinations—such as GCS with BUN or Lactate with SOFA—enhance predictive power by capturing multiple dimensions of critical illness. Overall, the EDA established that ICU mortality results from interconnected neurological, renal, hepatic, and metabolic dysfunctions, forming the empirical foundation for subsequent predictive modelling.

5. Modelling and Analysis

The modelling phase aimed to identify variables that most strongly predict in-hospital mortality across four intensive-care contexts: the Coronary Care Unit (ICU Type 1), the Cardiac Surgery Recovery Unit (ICU Type 2), the Medical ICU (ICU Type 3), and the Surgical ICU (ICU Type 4). Three supervised learning algorithms—logistic regression, random forest, and k-nearest neighbours (KNN)—were developed for each ICU and later compared on a combined dataset to examine shared and unit-specific predictors of mortality.

Each dataset was split into training and testing subsets using stratified sampling to maintain the ratio of survivors to non-survivors. Since mortality events were relatively rare, model evaluation focused not only on accuracy but also on precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC). Five-fold cross-validation ensured stability, and class-weight adjustments, threshold tuning, and synthetic minority oversampling (SMOTE) were applied to address imbalance. Examples of ROC-AUC comparison and Metrics Vs Threshold Graph to find optimal threshold for minority class prediction are included in the appendix, whereas confusion matrixes of best performing models for each ICU dataset are shown below.

ICU Type 1 – Coronary Care Unit

=== Logistic Regression (Optimized Threshold) ===

	precision	recall	f1-score	support
0	0.94	0.68	0.79	100
1	0.27	0.75	0.40	16
accuracy			0.69	116
macro avg	0.61	0.72	0.60	116
weighted avg	0.85	0.69	0.74	116

Feature	Coefficient	Odds_Ratio
BUN_avg2	0.718861	2.052095
AST_min2	0.640171	1.896805
GCS_max2	-0.569067	0.566054
NIMAPb_avg2	-0.425331	0.653553
SAPS.I	0.253529	1.288564
NISysABPb_avg2	-0.166429	0.846683
ALT_min2	-0.153526	0.857678
MechVentb_avg2	-0.125978	0.881634
SOFA	0.098598	1.103622
FiO2b_avg2	0.050475	1.051771

Figure: Confusion Matrix of Logistic Regression predicting In.hospital_death and coefficients of features selected

For cardiac-focused critical care, models were trained on neurological, hemodynamic, and biochemical features such as GCS, SOFA, SAPS-I, BUN, FiO_2, ALT, AST, and blood pressure readings. Logistic regression achieved 77–83 percent accuracy with an AUC of about 0.74, showing fair discrimination. Higher GCS values were protective, while elevated BUN and AST increased mortality risk. Random forest performed slightly better (AUC ≈ 0.77) and identified BUN, AST, and GCS as the most important predictors. KNN had high apparent accuracy but near-zero recall for deaths, reflecting bias from class imbalance. Overall, renal and hepatic dysfunction, neurological depression, and poor perfusion emerged as dominant mortality drivers.

ICU Type 2 – Cardiac Surgery Recovery Unit

```
=== Logistic Regression (Weighted, Optimal Threshold) ===
[[203   5]
 [  4   7]]
              precision    recall  f1-score   support

           0       0.98      0.98      0.98       208
           1       0.58      0.64      0.61        11

    accuracy                           0.96       219
   macro avg       0.78      0.81      0.79       219
weighted avg       0.96      0.96      0.96       219

ROC AUC Score: 0.9143
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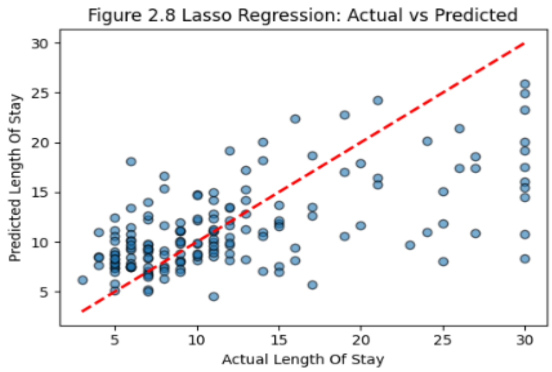


Figures: Confusion Matrix of Logistic Regression predicting In.hospital_death and Lasso Regression predicting Length of Stay

Among post-operative cardiac-surgery patients, mortality was low (4.9 percent) but clinically significant. GCS showed a strong negative correlation with death, while BUN correlated positively. Univariate logistic models confirmed that GCS alone discriminated well (AUC ≈ 0.81) but produced false positives, whereas BUN alone failed to detect deaths. A multivariable logistic regression including GCS, BUN, PaO_2, and $PaCO_2$ slightly improved recall but remained biased toward survival. Random forest achieved the best overall discrimination (AUC ≈ 0.77), while weighted logistic regression after SMOTE reached AUC 0.91 with balanced precision (≈ 0.58) and recall (≈ 0.64). These results highlight neurological impairment and renal dysfunction as early mortality indicators.

LASSO regression was also applied to predict Length of Stay (LOS) among survivors. The model achieved $R^2 \approx 0.37$, explaining about 37 percent of LOS variance with a mean absolute error of 4.16 days and RMSE of 5.77 days. Most prediction errors came from stays longer than 25 days, likely due to unmeasured clinical or logistical factors. Compared with linear and ridge regression, LASSO performed slightly better and highlighted the most influential predictors while reducing noise, providing an interpretable framework for understanding recovery duration.

ICU Type 3 – Medical ICU

```
=== Random Forest (Optimal Threshold) ===
[[274  28]
 [ 44  25]]
              precision    recall  f1-score   support

           0       0.86      0.91      0.88       302
           1       0.47      0.36      0.41        69

    accuracy                           0.81       371
   macro avg       0.67      0.63      0.65       371
weighted avg       0.79      0.81      0.80       371
```

Figure: Confusion Matrix of Random Forest predicting In.hospital_death

The Medical ICU had the highest mortality (18.6 percent) and the most complex pathophysiology, dominated by sepsis and multi-organ failure. Correlation analyses identified lactate, SOFA, GCS, and BUN as the strongest predictors, followed by bilirubin, albumin, and blood pressure. Logistic regression performed well for survivors but had low recall (0.20) for deaths. Random forest improved detection (AUC 0.72), while KNN trailed (0.66). Threshold tuning increased recall across models, with random forest achieving the best F1 balance. Feature-importance analysis confirmed that elevated lactate and SOFA, coupled with reduced GCS, were the clearest indicators of mortality. These findings align with clinical evidence that metabolic acidosis, renal dysfunction, and neurologic decline predict poor outcomes in medical ICU patients.

ICU Type 4 – Surgical ICU

```
=== Logistic Regression (Optimal Threshold) ===
Model accuracy: 0.78
Confusion matrix:
 [[285  75]
 [ 21  47]]

Classification report:
              precision    recall  f1-score   support

           0       0.93      0.79      0.86       360
           1       0.39      0.69      0.49        68

    accuracy                           0.78       428
   macro avg       0.66      0.74      0.68       428
weighted avg       0.84      0.78      0.80       428
```

Figure: Confusion Matrix of Logistic Regression predicting In.hospital_death

The Surgical ICU recorded a mortality rate of 14.5 percent. GCS again emerged as the most consistent predictor, with decreasing scores over time strongly linked to death. Logistic regression (AUC ≈ 0.81) outperformed other algorithms in identifying deaths, while KNN and random forest achieved higher overall accuracy but poor recall for mortality. After threshold optimization, recall improved substantially (KNN from 0.09 to 0.76; random forest from 0.12 to 0.66), though at the cost of increased false positives. Logistic regression provided the most balanced performance, confirming that monitoring changes in GCS over time offers valuable early warning for deterioration.

Combined and Cross-ICU Insights

```
=== Random Forest (Weighted, Optimal Threshold) ===
[[438 114]
 [ 28  49]]
              precision    recall  f1-score   support

           0       0.94      0.79      0.86       552
           1       0.30      0.64      0.41        77

    accuracy                           0.77       629
   macro avg       0.62      0.71      0.63       629
weighted avg       0.86      0.77      0.81       629

ROC AUC Score: 0.7692
```

Figure: Confusion Matrix of Random Forest predicting In.hospital_death

When all ICU data were merged, logistic regression and random forest achieved about 87 percent accuracy but modest recall at default thresholds. After reweighting and tuning, recall increased to 0.47 for logistic regression and 0.64 for random forest, with only slight precision loss. Random forest achieved the best balance, producing the highest recall and F1-score for deaths. Across models, neurological function (GCS) consistently ranked as the strongest predictor, followed by SOFA, SAPS-I, BUN, and lactate. Cardiovascular stability, reflected in mean arterial pressure, was protective, while elevated liver enzymes and poor oxygenation increased risk. Troponin and bilirubin correlated strongly with death in specific units but were limited by missing data, emphasizing the importance of data completeness.

Interpretation and Implications

Overall, ICU mortality was influenced by intertwined neurological, hemodynamic, and metabolic disturbances rather than any single variable. Logistic regression provided interpretability, random forest captured complex interactions, and KNN identified clusters of patients with similar outcomes. Random forest's higher AUC and F1-scores suggest that mortality risk is multidimensional, while logistic regression remains valuable for clinical transparency. Addressing class imbalance through weighting and oversampling was essential to improve death detection, highlighting that accuracy alone is insufficient when the minority class carries the highest clinical cost.

Common predictors such as age, GCS, lactate, BUN, and mean arterial pressure reflect core physiological processes underlying critical illness: perfusion, metabolism, and neurological integrity. Each ICU also displayed unique risk signatures tied to its clinical focus, including Troponin in coronary care, surgical duration and postoperative lactate in cardiac recovery, and temperature trends in surgical trauma cases. Together, these findings demonstrate that combining interpretable statistical models with ensemble learning produces clinically meaningful and operationally useful tools. Embedding such predictive frameworks into ICU information systems could enhance early-warning capabilities, guide triage and resource allocation, and ultimately help reduce preventable deaths in critical care.

6. Conclusion

This study aimed to identify the key factors influencing patient mortality across four ICU types—Coronary Care (Type 1), Cardiac Surgery Recovery (Type 2), Medical (Type 3), and Surgical (Type 4)—and to evaluate predictive models for capturing these relationships. Using the PhysioNet/CinC Challenge 2012 dataset of over 4,000 ICU admissions and 42 clinical variables, the analysis showed that while mortality determinants differ across ICU types due to patient and treatment variations, several physiological markers consistently predict mortality across all settings.

Neurological status (GCS), organ dysfunction scores (SOFA and SAPS-I), and renal-metabolic indicators (BUN and Lactate) emerged as the most influential predictors of death. These reflect the interplay between neurological deterioration, multi-organ failure, and metabolic imbalance—core drivers of critical illness. Some predictors were ICU-specific: cardiac-related variables (AST, ALT, FiO_2) were significant in the Coronary Care Unit, while post-operative markers were more relevant in the Surgical and Cardiac Recovery Units, emphasizing the distinct clinical focus of each ICU type.

In predictive modelling, Logistic Regression consistently achieved the best mortality discrimination in specialized ICUs (Types 1, 2, and 4) due to its interpretability and stability. Random Forest outperformed other models in more complex or mixed-patient settings (Medical ICU and combined analysis), capturing nonlinear and interaction effects between variables. Lasso Regression was most effective in predicting length of stay, particularly in short-term recovery contexts, although its performance was limited by incomplete data.

Overall, the results show that ICU mortality is influenced by multiple interdependent systems rather than isolated physiological markers. The consistent importance of GCS, SOFA, SAPS-I, BUN, and Lactate across models highlights their reliability as early warning indicators for clinicians. Predictive modelling, therefore, offers valuable support for early intervention, triage, and resource allocation, ultimately improving patient outcomes and ICU management efficiency.

For future work, enhancing data completeness is crucial. Continuous collection of patient variables throughout ICU stays—rather than only at admission—would reduce missing data, improve model accuracy, and strengthen both mortality and length-of-stay predictions, particularly for models like Lasso Regression.

In summary, while mortality patterns differ across ICU types, neurological function, organ failure scores, and renal-metabolic balance remain the most universal and clinically significant predictors. Combining interpretable statistical models with ensemble learning provides a strong, practical foundation for improving ICU decision-making and benchmarking quality of care.

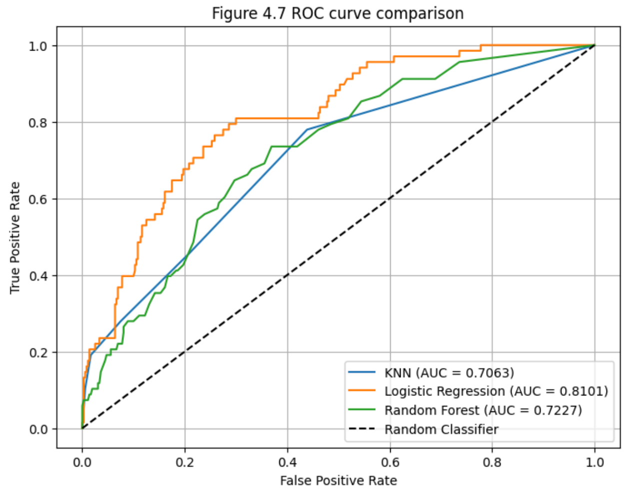
Appendix

Group Project Assessment for ADS1002						
Studio: Malaysia						
Group: ICU_3						
		Lee Jun Siang, 3574567	**Nafil Alfritanto, 35543167**	**Sharon Lim Jia En, 35887311**	**Leng Jaek-Hxiang, 35897317**	**Bong Min Ge, 33608458**
Coordination	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	20	20	20	20	20
Research	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	20	20	20	20	20
Coding	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	20	20	20	20	20
Report Writing	The sum of this row should be 100. The entry should reflect the group member's percentage	20	20	20	20	20

	contribution to this project task.					
Presentation Preparation	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	20	20	20	20	20
Attendance at Final Presentation	This entry should be 1 (present) or 0 (absent)	1	1	1	1	1

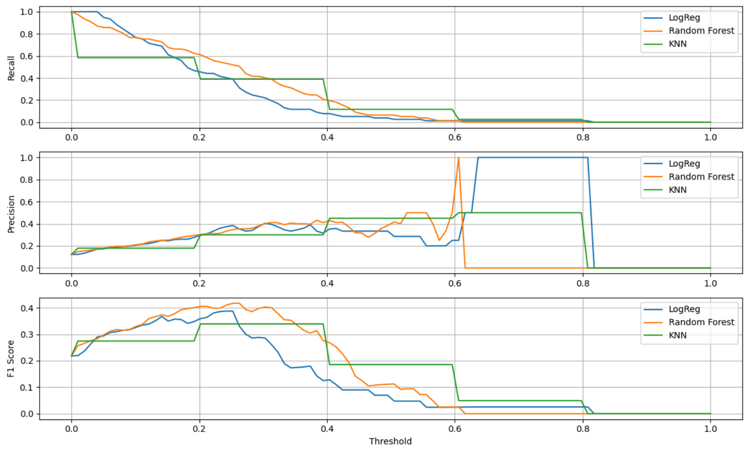
Acknowledgement of AI use: No AI was used for this assignment.

Example: ROC-AUC comparison to determine best performing model:



Example: Metrics Vs Threshold Graph to find optimal threshold for minority class prediction:

Figure 5.3: Metrics vs. Threshold Comparison (LogReg, RF, KNN)



Figures of confusion matrixes & classification reports of each subquestion:

Subquestion 1(ICU type 1):

```
=== Logistic Regression (Optimized Threshold) ===
              precision    recall  f1-score   support

           0       0.94      0.68      0.79       100
           1       0.27      0.75      0.40        16

    accuracy                           0.69       116
   macro avg       0.61      0.72      0.60       116
weighted avg       0.85      0.69      0.74       116

Best F1: 0.4000, Threshold: 0.08, ROC AUC: 0.7344

=== Random Forest (Optimized Threshold) ===
              precision    recall  f1-score   support

           0       0.91      0.82      0.86       100
           1       0.31      0.50      0.38        16

    accuracy                           0.78       116
   macro avg       0.61      0.66      0.62       116
weighted avg       0.83      0.78      0.80       116

Best F1: 0.3810, Threshold: 0.22, ROC AUC: 0.7103

=== K-Nearest Neighbors (Optimized Threshold, K=20) ===
              precision    recall  f1-score   support

           0       0.92      0.80      0.86       100
           1       0.31      0.56      0.40        16

    accuracy                           0.77       116
   macro avg       0.61      0.68      0.63       116
weighted avg       0.84      0.77      0.79       116

Best F1: 0.4000, Threshold: 0.16, ROC AUC: 0.7178
```

Subquestion 2(ICU type 2):

```
=== Logistic Regression (Weighted, Optimal Threshold) ===
[[203   5]
 [  4   7]]
              precision    recall  f1-score   support

           0       0.98      0.98      0.98       208
           1       0.58      0.64      0.61        11

    accuracy                           0.96       219
   macro avg       0.78      0.81      0.79       219
weighted avg       0.96      0.96      0.96       219

ROC AUC Score: 0.9143

=== Random Forest (Weighted, Optimal Threshold) ===
[[208   0]
 [  8   3]]
              precision    recall  f1-score   support

           0       0.96      1.00      0.98       208
           1       1.00      0.27      0.43        11

    accuracy                           0.96       219
   macro avg       0.98      0.64      0.70       219
weighted avg       0.96      0.96      0.95       219

ROC AUC Score: 0.8184

=== KNN (SMOTE + Distance Weights, Optimal Threshold) ===
[[195  13]
 [  5   6]]
              precision    recall  f1-score   support

           0       0.97      0.94      0.96       208
           1       0.32      0.55      0.40        11

    accuracy                           0.92       219
   macro avg       0.65      0.74      0.68       219
weighted avg       0.94      0.92      0.93       219

ROC AUC Score: 0.8295
```

Subquestion 3(ICU type 3):

```
=== Logistic Regression (Optimal Threshold) ===
[[272  30]
 [ 48  21]]
              precision    recall  f1-score   support

           0       0.85      0.90      0.87       302
           1       0.41      0.30      0.35        69

    accuracy                           0.79       371
   macro avg       0.63      0.60      0.61       371
weighted avg       0.77      0.79      0.78       371


=== Random Forest (Optimal Threshold) ===
[[274  28]
 [ 44  25]]
              precision    recall  f1-score   support

           0       0.86      0.91      0.88       302
           1       0.47      0.36      0.41        69

    accuracy                           0.81       371
   macro avg       0.67      0.63      0.65       371
weighted avg       0.79      0.81      0.80       371


=== KNN (Optimal Threshold) ===
[[216  86]
 [ 37  32]]
              precision    recall  f1-score   support

           0       0.85      0.72      0.78       302
           1       0.27      0.46      0.34        69

    accuracy                           0.67       371
   macro avg       0.56      0.59      0.56       371
weighted avg       0.75      0.67      0.70       371
```

Subquestion 4(ICU type 4):

```
=== KNN (Optimal Threshold) ===
Model accuracy: 0.72
Confusion matrix:
 [[257 103]
 [ 16  52]]

Classification report:
              precision    recall  f1-score   support

           0       0.94      0.71      0.81       360
           1       0.34      0.76      0.47        68

    accuracy                           0.72       428
   macro avg       0.64      0.74      0.64       428
weighted avg       0.85      0.72      0.76       428


=== Logistic Regression (Optimal Threshold) ===
Model accuracy: 0.78
Confusion matrix:
 [[285  75]
 [ 21  47]]

Classification report:
              precision    recall  f1-score   support

           0       0.93      0.79      0.86       360
           1       0.39      0.69      0.49        68

    accuracy                           0.78       428
   macro avg       0.66      0.74      0.68       428
weighted avg       0.84      0.78      0.80       428


=== Random Forest (Optimal Threshold) ===
Model accuracy: 0.68
Confusion matrix:
 [[244 116]
 [ 23  45]]

Classification report:
              precision    recall  f1-score   support

           0       0.91      0.68      0.78       360
           1       0.28      0.66      0.39        68

    accuracy                           0.68       428
   macro avg       0.60      0.67      0.59       428
weighted avg       0.81      0.68      0.72       428
```

Subquestion 5(All ICU types):

```
=== Logistic Regression (Weighted, Optimal Threshold) ===
[[482  70]
 [ 41  36]]
              precision    recall  f1-score   support

           0       0.92      0.87      0.90       552
           1       0.34      0.47      0.39        77

    accuracy                           0.82       629
   macro avg       0.63      0.67      0.65       629
weighted avg       0.85      0.82      0.84       629

ROC AUC Score: 0.7612

=== Random Forest (Weighted, Optimal Threshold) ===
[[438 114]
 [ 28  49]]
              precision    recall  f1-score   support

           0       0.94      0.79      0.86       552
           1       0.30      0.64      0.41        77

    accuracy                           0.77       629
   macro avg       0.62      0.71      0.63       629
weighted avg       0.86      0.77      0.81       629

ROC AUC Score: 0.7692

=== KNN (SMOTE + Distance Weights, Optimal Threshold) ===
[[487  65]
 [ 48  29]]
              precision    recall  f1-score   support

           0       0.91      0.88      0.90       552
           1       0.31      0.38      0.34        77

    accuracy                           0.82       629
   macro avg       0.61      0.63      0.62       629
weighted avg       0.84      0.82      0.83       629

ROC AUC Score: 0.6524
```